

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures

Code:

```
geya@LenovoL480: ~/OS/Skil  ×  +  v
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main() {
    char command[100];

    printf("Enter command: ");
    fgets(command, sizeof(command), stdin);

    command[strcspn(command, "\n")] = '\0';

    if (strlen(command) == 0) {
        printf("Error: Empty command\n");
    }
    else if (strcmp(command, "ls") == 0) {
        printf("Valid command: ls\n");
    }
    else if (strcmp(command, "pwd") == 0) {
        printf("Valid command: pwd\n");
    }
    else if (strcmp(command, "exit") == 0) {
        printf("Valid command: exit\n");
    }
    else {
        printf("Unknown command\n");
    }

    return 0;
}
```

Output:

```
geya@LenovoL480:~/OS/Skill/Week 2$ gcc task4.c -o task4
geya@LenovoL480:~/OS/Skill/Week 2$ ./task4
Enter command: ls
Valid command: ls
geya@LenovoL480:~/OS/Skill/Week 2$ ./task4
Enter command:
Error: Empty command
geya@LenovoL480:~/OS/Skill/Week 2$ |
```